

PRML Spring 2026 Homework 4: Transformer Models

Yifan Ding

Code: <https://github.com/douhenhuidyf/PRML-Spring-2026/tree/main/hw4>

Email: 23373493@buaa.edu.cn

May 15, 2026

1 Introduction

1.1 Problem Statement

Read and reproduce the classic 2017 paper: "Attention is All You Need"[2]. Then, through experiments, try to understand the necessity of each module in the Transformer. You can choose to study any of the following situations:

1. Positional Encoding: Compare the original method with simple absolute encoding or other forms of positional encoding, and discuss the core role of positional encoding.
2. Necessity of Q, K, V: If we only use Q and K (merging K and V into one matrix), what is the impact on the algorithm?
3. If the ResNet structure is not adopted in the module, what is the impact on the algorithm?
4. After introducing positional encoding, reconstruct a network using CNN and compare it with the self-attention mechanism of the Transformer.
5. Which direction can be improved in the classic Transformer module? (Not directly citing other published improvement methods) Propose independent ideas and practice to verify the feasibility of your ideas.

1.2 Problem Analysis

The Transformer model, introduced by Vaswani et al. in 2017, was originally designed for natural language processing tasks, especially machine translation. Thus, we will focus on the performance of the Transformer in translation tasks in this homework. As training a Transformer from scratch can be computationally expensive, we will use a smaller Chinese to English translation dataset and a smaller scale configuration for our experiments.

1.3 Structure of this paper

The rest of this paper is organized as follows. In ??, we introduce the evaluation metrics and the Transformer variants considered in this study. ?? reports the experimental setup, final results, and analysis of training dynamics. We conclude in ?? with key findings and practical implications.

2 Methodology

2.1 Metrics

To evaluate the performance of the baseline Transformer model and its variants, we use two standard machine translation metrics: BLEU and chrF. Both are computed at the corpus level. BLEU (Bilingual Evaluation Understudy) measures n -gram precision with a brevity penalty to discourage overly short translations.

It is defined as:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right),$$

where p_n is the modified n -gram precision, w_n is the weight (uniform in our experiments), and BP is the brevity penalty.

chrF measures character n -gram overlap using an F-score, which is more sensitive to morphological and subword differences. It is defined as:

$$\text{chrF}_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}},$$

where precision and recall are computed over character n -grams, and we use $\beta = 2$ to emphasize recall.

2.2 Transformer Architecture

The baseline model follows the standard encoder-decoder Transformer. Each encoder layer consists of a multi-head self-attention sublayer and a position-wise feed-forward network (FFN). Each decoder layer contains masked self-attention, encoder-decoder attention, and a FFN. Residual connections and LayerNorm are used around each sublayer. We use the same tokenization and vocabulary across all variants for a fair comparison.

2.3 Variants of Transformer

We study three architectural variants to understand the role of residual paths, normalization placement, and feed-forward networks.

2.3.1 No Residual Connections

We remove the residual connections in both the attention and feed-forward sublayers while keeping all other settings unchanged. This variant tests whether the shortcut paths are necessary for stable optimization and effective gradient flow in deep Transformer stacks.

Theoretically, without residual connections, the model may struggle to learn identity mappings, which can lead to vanishing gradients and hinder training convergence. We will empirically verify this by comparing the training dynamics and final performance of this variant against the baseline Transformer.

2.3.2 Post-Norm Transformer

We compare the common pre-norm design with a post-norm setting which is the original design in the "Attention is All You Need!" paper. In the post-norm design, LayerNorm is applied after the residual connection, rather than before it as in the pre-norm design. This means that the output of each sublayer is normalized after adding the input (residual) to the output of the sublayer, rather than normalizing the sublayer output before adding the residual. The post-norm structure can lead to different training dynamics and stability properties compared to pre-norm, and we will empirically compare their performance on our translation task. The structure is illustrated in ??.

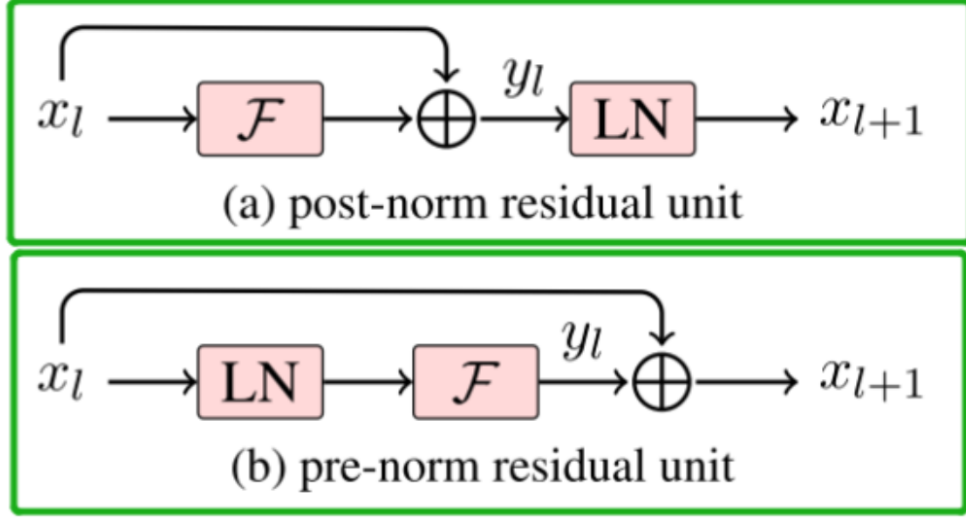


Figure 1: Pre-norm and post-norm structures for a Transformer sublayer.

2.3.3 MoE with Linear Routing

We introduce a lightweight Mixture-of-Experts (MoE) layer in the feed-forward block. The router uses a simple linear projection to compute expert scores, and tokens are sent to top- k experts without additional routing networks.

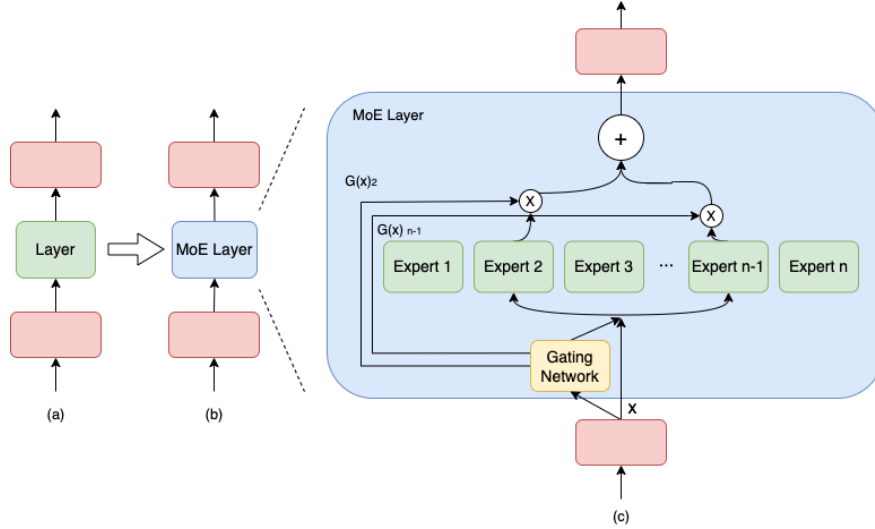


Figure 2: MoE feed-forward block with a linear routing projection.

MoEs can increase model capacity without a proportional increase in computational cost, as only a subset of experts is activated for each token. We will evaluate whether this simple MoE design can improve translation performance compared to the standard feed-forward network in the Transformer.

3 Experiments

3.1 Implement Details

The baseline Transformer has 8 transformer blocks for encoder and decoder layers with a model dimension of 1536, 6 attention heads, and a feed-forward dimension of 4095. The MoE variant uses 4 experts in the feed-forward block with top-1 routing. We evaluate on the validation set every 50 steps and compute BLEU and chrF using SacreBLEU. All models are trained with the same AdamW optimizer, OneCycleLR learning rate schedule, batch size of 128, and training steps of 600. We use a single NVIDIA A600 48GB GPU for training, and the code is implemented in PyTorch.

3.2 Dataset

We will use the translation2019zh[3] Chinese to English translation dataset, which contains around 5200k sentence pairs for training and 40k sentence pairs for validation. Here, we will use a tokenizer[1] to preprocess the dataset, and set the maximum sequence length to 192 for english sentences and 128 for chinese sentences.

3.3 Experiment Result

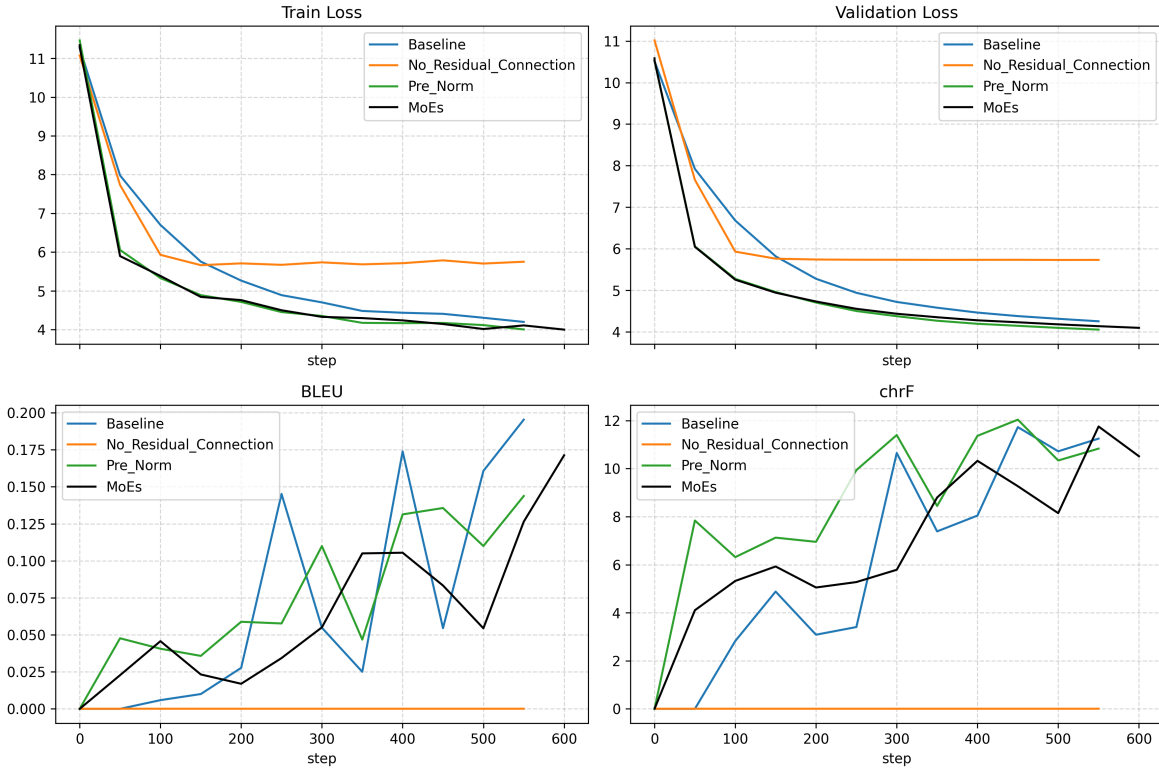


Figure 3: Training curves of loss, BLEU, and chrF for each model variant.

?? shows clear differences in training dynamics. The no-residual variant quickly plateaus in both train and validation loss, and its BLEU/chrF stay at zero, indicating optimization failure without shortcut paths. The pre-norm model reduces training oscillations compared to post-norm (baseline), with smoother loss curves and a more monotonic rise of chrF, suggesting improved stability. However, its final BLEU is lower than the baseline, implying that stability alone does not guarantee better translation quality in this setup. The MoE variant achieves the lowest training loss and competitive validation loss, but its BLEU/chrF do not

surpass the baseline, which suggests that a simple linear router increases capacity without reliably improving sequence-level accuracy under the current data and training budget.

As shown in ??, the baseline Transformer achieves the highest BLEU and chrF scores, while the no-residual variant fails to learn effectively. The pre-norm variant offers a more stable training process but does not outperform the baseline in final translation quality. The MoE variant reduces training loss but does not translate into higher BLEU or chrF, indicating that while MoEs can increase model capacity without proportional increase in computational cost, routing quality or training scale may be the bottleneck for realizing its potential benefits in this setting.

Table 1: Final results on the validation set.

Model	Train Loss	Val Loss	BLEU	chrF
Baseline	4.1966	4.2527	0.1954	11.2479
Pre-Norm	4.0024	4.0522	0.1438	10.8320
No Residual	5.7507	5.7307	0.0000	0.0000
MoE (Linear Router)	<i>textbf</i> 3.9981	4.0947	0.1713	10.5144

4 Conclusions

We reproduced a small-scale Transformer for Chinese-to-English translation and compared three variants. Residual connections are essential for deep neural network optimization; removing them causes training to collapse. Pre-norm improves stability and reduces oscillation, yet does not outperform the baseline on BLEU. The lightweight MoE improves training loss but does not translate into higher BLEU or chrF, indicating that routing quality or training scale and scale of training steps may be the bottleneck. Overall, the baseline remains the strongest in final translation quality, while pre-norm offers a more stable optimization profile.

References

- [1] Jörg Tiedemann and Santhosh Thottingal. “OPUS-MT — Building open translation services for the World”. In: *Proceedings of the 22nd Annual Conferenec of the European Association for Machine Translation (EAMT)*. Lisbon, Portugal, 2020.
- [2] Ashish Vaswani et al. *Attention Is All You Need*. 2023. arXiv: 1706.03762 [cs.CL]. URL: <https://arxiv.org/abs/1706.03762>.
- [3] Bright Xu. *NLP Chinese Corpus: Large Scale Chinese Corpus for NLP*. Version 1.0. Sept. 2019. DOI: 10.5281/zenodo.3402023. URL: <https://doi.org/10.5281/zenodo.3402023>.